Some problems in field theory

Aditya Padiyar

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1. Give an example of a degree 4 extension with exactly one intermediate subfield

   Solution Consider the fields $\mathbb{Q} \subset \mathbb{Q}(\alpha^2) \subset \mathbb{Q}(\alpha)$, where $\alpha > 0$ is such that $\alpha^4 - 2 = 0$

2. Give an example of a degree 4 extension with infinitely many intermediate subfields

   Solution Consider the fields $F \subset F(\alpha + c\beta) \subset \frac{R}{I}$ where $F = \mathbb{F}_2(x, y)$, $R = F[s, t], I = (s^2 - x, t^2 - y), \alpha = s + I, \beta = t + I, c \in F$

3. Let $\alpha = \sqrt{2}$ and $\beta > 0$ be such that $\beta^3 - 5 = 0$. Show that $[\mathbb{Q}(\alpha + \beta) : \mathbb{Q}] = 6$

   Solution Use the fact that a $\mathbb{Q}$-basis for $\mathbb{Q}(\alpha, \beta)$ is given by

   $$\{1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2\}$$

4. Suppose we have $F \subset R \subset K$, where F, K are fields, R is a ring, and K is algebraic over F. Show that R is a field.

   Solution Let $r \neq 0 \in R$. The map M from $F[r] \to F[r]$ given by $x \to rx$ is F-linear and injective. Since $F[r]$ is finite dimensional over F, M must be surjective as well.

5. Suppose $F \subset K$ is a finite extension and $\alpha_1 \in K$. Consider the F-linear map M from $K \to K$ given by $x \to \alpha_1 x$. Find the trace, determinant and characteristic polynomial of M

   Solution We let $K = F(\alpha_1, \ldots, \alpha_n)$, so that a basis of K is given by $\alpha_1^{j_1}\alpha_2^{j_2}\ldots\alpha_n^{j_n}$. Let $p(x) \in F[x]$ be the irreducible polynomial of α and C its companion matrix. The matrix of M is then $\text{diag}(C, C, \ldots, C)$.

6. Let $\alpha = \cos(\frac{2\pi}{7})$. Show that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$

   Solution Let $w = e^{i\frac{2\pi}{7}}$. Then we have $w^2 - 2\alpha w + 1 = 0$

7. Suppose $f(x), g(x)$ are coprime in $F[x]$ with $\deg f < \deg g = n$. Let $\alpha = \frac{f(x)}{g(x)} \in F(x)$. Show that $[F(x) : F(\alpha)] = n$

   Solution Use Gauss's lemma

8. Suppose that every polynomial in $F[x]$ has a root in a separable algebraic extension K of F. Show that K is algebraically closed.

   Solution Suppose $f(x) = (x - \alpha_1) \dots (x - \alpha_n) \in F[x]$ where $\alpha_i \in \overline{K}$. By the primitive element theorem we have $F(\alpha_1, \dots, \alpha_n) = F(\beta)$. Let $p(x) \in F[x]$ be the minimal polynomial of β over F. By the hypothesis, $p(\alpha) = 0$ with $\alpha \in K$. Hence we have an isomorphism $\phi : F(\beta) \to F(\alpha)$ that fixes F with $\phi(\beta) = \alpha$. Hence $\alpha_i \in K$